Year 11 Set 4 · Foundation · Lesson 2

Ratio problems

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Today I will learn to...

Goals

1. Share an amount in a given ratio

2. Solve ratio problems given one part or the difference

Do now: retrieval practice

4 mins

1. Increase £80 by 25%

   £100

2. Simplify 12 : 18

   2 : 3

3. Write 2/5 as a decimal

   0.4

4. Find 15% of 200

   30

Objective 1

Share an amount in a given ratio

Key idea

Learn

!

Add the parts, find one part, then multiply. Check the shares add back to the total.

Objective 1: Share an amount in a given ratio

I do: example 1

I do

Share £60 in the ratio 2 : 3.

1. 2 + 3 = 5 parts
2. 60 ÷ 5 = 12 per part
3. 2 × 12 = £24, 3 × 12 = £36

I do: example 2

I do

Share 45 sweets in the ratio 4 : 5.

1. 9 parts
2. $45 \div 9 = 5$
3. 20 and 25

I do: example 3

I do

Share 120 in the ratio 3 : 2 : 1.

1. 6 parts
2. 120 ÷ 6 = 20
3. 60, 40 and 20

Mini whiteboards

We do

1. Share 40 in the ratio 3 : 5

Answer: 15 and 25

2. Share 70 in the ratio 1 : 6

Answer: 10 and 60

3. Share 36 in the ratio 5 : 4

Answer: 20 and 16

4. Share 60 in the ratio 1 : 2 : 3

Answer: 10, 20 and 30

You do: choose your challenge

You do

Bronze

Share £35 in the ratio 3 : 4.

Silver

Share 90 g in the ratio 2 : 3 : 5.

Gold

Sam and Tom share £150 in the ratio 7 : 3. How much more does Sam get than Tom?

Platinum

Red : blue beads = 5 : 3. There are 64 beads. 4 red beads are removed. Find the new ratio of red to blue in its simplest form.

You do: answers

Check

Bronze

£15 and £20

Silver

18 g, 27 g and 45 g

Gold

1 part = £15. Sam £105, Tom £45. £60 more

Platinum

Red 40, blue 24. 36 : 24 = 3 : 2

If you finish early

Extra

Bronze

Share £24 in the ratio 1 : 3.

£6 and £18

Silver

Share 84 kg in the ratio 2 : 5.

24 kg and 60 kg

Gold

Share £200 in the ratio 3 : 3 : 4.

£60, £60 and £80

Platinum

Ali, Bea and Cal share £180 in the ratio 2 : 3 : 4. Cal gives half of his share to Ali. How much does Ali have now?

£40, £60, £80. Ali has 40 + 40 = £80

Exam question 1

2 marks

Amy and Ben share £72 in the ratio 5 : 3. How much does Amy get?

Space for working

Mark scheme: 72 ÷ 8 = 9, 5 × 9 = £45

Exam question 2

2 marks

Paint is mixed red : white = 2 : 7. There are 36 litres in total. How much is white?

Space for working

Mark scheme: 36 ÷ 9 = 4, 7 × 4 = 28 litres

Objective 1 achieved

Share an amount in a given ratio



Objective 2

Solve ratio problems given one part or the difference

✓ Achieved: Share an amount in a given ratio

Key idea

Learn

!

Work out what ONE part is worth from the information you are given, then use it.

Objective 2: Solve ratio problems given one part or the difference

I do: example 1

I do

Boys : girls = 3 : 5. There are 12 boys. How many girls?

1. 3 parts = 12, so 1 part = 4
2. Girls = 5 × 4 = 20

I do: example 2

I do

Two numbers are in the ratio 2 : 5 and their difference is 18. Find them.

1. Difference = 5 − 2 = 3 parts = 18
2. 1 part = 6
3. 12 and 30

I do: example 3

I do

A : B = 4 : 1. A is 28. Find the total.

1. 4 parts = 28, 1 part = 7
2. B = 7, total = 35

Mini whiteboards

We do

1. Ratio 3 : 4, smaller is 15. Larger?

Answer: 20

2. Ratio 1 : 3, difference 10. The two numbers?

Answer: 5 and 15

3. Ratio 2 : 5, larger is 35. Smaller?

Answer: 14

4. Ratio 3 : 7, difference 20. Total?

Answer: 1 part = 5, total 50

You do: choose your challenge

You do

Bronze

Ratio 2 : 3. The smaller amount is 8. Find the larger.

Silver

Ratio 4 : 7. The difference is 15. Find the total.

Gold

Ann : Ben's ages = 3 : 5. Ben is 12 years older. How old is Ann?

Platinum

Biscuit mix uses flour : sugar : butter = 5 : 2 : 3. Tia has 400 g flour, 200 g sugar and 210 g butter. What is the most mix she can make?

You do: answers

Check

Bronze

12

Silver

1 part = 5, total = 55

Gold

2 parts = 12, 1 part = 6, Ann is 18

Platinum

Butter runs out first: 1 part = 70 g. Total = 10 × 70 = 700 g

If you finish early

Extra

Bronze

Ratio 1 : 4. The smaller amount is 6. Find the larger.

24

Silver

Boys : girls = 5 : 3. There are 15 girls. How many boys?

25

Gold

Ratio 5 : 8. The difference is 21. Find both amounts.

1 part = 7: 35 and 56

Platinum

Paint is mixed red : white = 2 : 7. Jo has 1.2 litres of red and 4 litres of white. What is the most paint she can make?

White runs out first: 1 part = 4 ÷ 7. Total = 9 × 4/7 = 5.14 litres

Exam question 1

3 marks

Cats : dogs = 2 : 7. There are 25 more dogs than cats. How many cats are there?

Space for working

Mark scheme: 5 parts = 25, 1 part = 5, cats = 10

Exam question 2

2 marks

A recipe for 4 people uses 300 g of flour. How much flour is needed for 10 people?

Space for working

Mark scheme: 300 ÷ 4 × 10 = 750 g

Objective 2 achieved

Solve ratio problems given one part or the difference



Real life

Apply

Concrete is cement : sand : gravel = 1 : 2 : 4. A builder uses 10 kg of cement. How much concrete does he make?

7 parts × 10 = 70 kg

Extension: stretch yourself

Stretch

1. x : y = 3 : 4 and y : z = 2 : 5. Find x : z.

x : y : z = 3 : 4 : 10, so x : z = 3 : 10

2. A : B = 2 : 3. After 10 is added to A the ratio is 1 : 1. Find A and B.

2k + 10 = 3k → k = 10, A = 20, B = 30

Exit ticket: exam questions

Exit

1. Write 45 : 60 in its simplest form. **(1 mark)**

2. Share £120 in the ratio 1 : 2 : 5. **(2 marks)**

3. In a club, boys : girls = 4 : 5. There are 12 more girls than boys. How many people are in the club? **(3 marks)**

Exit ticket: mark scheme

Mark it

Q1 3 : 4 [1]

Q2 £15, £30, £75 [2]

Q3 1 part = 12, 9 parts = 108 [3]

Total: ____ / 6

Today I achieved

- ✔ **Share an amount in a given ratio**
- ✔ **Solve ratio problems given one part or the difference**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework: exam questions

Due next lesson

1. Write 24 : 36 in its simplest form. [1]

2. Share £84 in the ratio 5 : 2. [2]

3. Cats : dogs = 3 : 8. There are 24 dogs. How many cats are there? [2]

4. A recipe for 6 people uses 450 g of mince. How much mince is needed for 10 people? [2]

5. Ali, Bea and Cal share money in the ratio 2 : 3 : 4. Cal gets £30 more than Ali. How much money is shared altogether? [3]

Extra practice: Maths Genie → Edexcel question bank → Ratio